QUADRATIC EXCLUSION LAWS FOR CONSECUTIVE ARTIN PRIMES IN ARBITRARY BASES

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ABSTRACT. Say that an odd prime p is *Artin base a* if a is a primitive root modulo p. (We restrict to odd primes throughout: modulo 2 every odd integer trivially has full multiplicative order, and the quadratic obstruction below fails there.) Membership in A_a forces the quadratic character value $\chi_a(p) = -1$, and since χ_a has conductor f, this necessary condition depends only on $p \bmod f$. We determine, for every non-square base a, the exact set of gap classes on which the resulting *quadratic obstruction* forbids two primes from both being Artin base a. Two mechanisms produce such *exclusion classes*. The first is *reversal*: $\chi_a(q) = -\chi_a(p)$ for all admissible $p, q = p + g$, so one of the two primes is a quadratic residue base a; we classify the reversing classes completely via the factorization of the discriminant of $\mathbb{Q}(\sqrt{a})$ into prime discriminants. The second, operating when no reversing class exists, is *inadmissibility*. Its analysis rests on an exact count, and we prove the general identity $4N_{--}(g) = T(g) - A(g) - B(g) + S(g)$ valid for every modulus, prime or composite, from which the prime-conductor evaluation $N_{--}(g) = (d - 3 + 2\chi(g))/4$ for $g \not\equiv 0 \pmod d$ (and $N_{--}(0) = (d-1)/2$) follows as a special case. The resulting complete dichotomy is: base a admits a quadratic exclusion class if and only if d is even, $d \equiv 3 \pmod 4$, $3 \mid d$, or $d = 5$. We emphasise throughout that the quadratic obstruction is *necessary but not sufficient* for Artin status, so an exclusion class is a proof of impossibility, while the absence of one is not a proof of possibility. The classification is verified by exhaustive computation for all non-square $2 \le a \le 80$, and the exclusion laws hold without exception across 84,981,870 base–pair incidences below 10^9. The counting identity, together with the $d = 5$ and $d = 13$ cases of the prime-conductor count, is additionally machine-checked in Lean 4.

We then measure, for eleven bases, the consecutive-pair Artin correlation $\delta(a)$ over all 50,847,531 consecutive prime pairs below 10^9 with smaller member at least 5. The exclusion weight w correlates with $|\delta|$ at $r = 0.646$, but this association is confounded with the conductor: $r(\log f, |\delta|) = -0.957$, the partial correlation of w given $\log f$ is only 0.136, and the three bases with no exclusion classes still carry $|\delta|$ up to 0.043. Restricting δ to gap classes outside the exclusion set yields a positive value for all eight bases that possess such classes. Among the quantities compared, the conductor of χ_a is the stronger descriptive predictor of the global correlation; we do not claim it is the operative mechanism.

1. INTRODUCTION

Artin's conjecture on primitive roots asserts that for a non-square integer $a \neq -1$ the set of primes p for which a is a primitive root modulo p has a positive density, given by an explicit rational multiple of Artin's constant. It is known under GRH by Hooley [8]. Unconditionally, and for the weaker assertion of infinitude rather than positive density, Heath-Brown [7], building on Gupta–Murty [5], proved that all but at most two prime bases are primitive roots for infinitely many primes; in particular, of any three distinct primes, at least one is a primitive root for infinitely many p. This infinitude statement should be kept separate from Hooley's conditional

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density theorem. See Moree's survey [12] for the extensive literature, and [2, 13] for recent uniform versions.

Write

$$\mathrm{A}_a = \{\, p \text{ an odd prime} : a \text{ is a primitive root mod } p \,\},$$

and call $p \in \mathrm{A}_a$ an *Artin prime base* a. The restriction to odd primes is not cosmetic. Modulo 2 the group $(\mathbb{Z}/2\mathbb{Z})^\times$ is trivial, so every odd a vacuously has full order, while $\chi_a(2)$ need not be -1; taking $a = 33$, the primes 2 and 13 both have 33 as a primitive root even though their gap 11 lies in a reversing class modulo the conductor 33. Every statement below therefore concerns odd primes only, as the hypotheses of Lemma 5 and Theorem 8 already record. All results above concern a *single* prime at a time. The behaviour of Artin primes in *pairs* has been studied much less. Garcia, Kahoro and Luca [3] (see also [4]) investigated primitive-root bias for twin primes; Pollack [14] showed under GRH that for a fixed admissible base there are infinitely many bounded-gap runs of primes all having that base as a primitive root; Baker–Pollack [1] proved unconditionally that there are strings of consecutive primes each having *some* primitive root from a large prescribed set, where the root may depend on the prime.

The starting point of this paper is an elementary but apparently unrecorded observation. Since a quadratic residue cannot generate the cyclic group $(\mathbb{Z}/p\mathbb{Z})^\times$ of even order $p-1$, membership in A_a forces $\chi_a(p) = -1$, where χ_a is the quadratic character attached to $\mathbb{Q}(\sqrt{a})$. Because χ_a is a character modulo the conductor f of that field, the value $\chi_a(p)$ depends only on $p \bmod f$. Consequently, if a shift g has the property that it *reverses* χ_a on every admissible residue class modulo f, then for any two primes p and $q = p + g'$ with $g' \equiv g \pmod f$, *neither dividing* a, we get $\chi_a(q) = -\chi_a(p)$, so at least one of $\chi_a(p), \chi_a(q)$ equals $+1$ — and that prime is not Artin base a. A prime dividing a is not Artin base a either, since then a is not even a unit; so two odd primes at such a gap can never *both* be Artin base a. We call this a *quadratic exclusion law* (Theorem 8).

What this paper does and does not classify. It is essential to be precise about the scope of the classification, and we state the limitation once, plainly, so that no result below is read as more than it is. The condition $\chi_a(p) = -1$ is *necessary* for $p \in \mathrm{A}_a$ but far from *sufficient*: a must also be a non-ℓ-th-power residue modulo p for every odd prime $\ell \mid p-1$. What we classify completely is the set of gap classes on which the *quadratic* obstruction alone forbids a doubly-Artin pair (Definition 7). Therefore:

- when a gap class *is* a quadratic exclusion class, no pair of *odd* primes at that gap can both be Artin base a — an unconditional theorem;
- when a gap class is *not* a quadratic exclusion class, we conclude nothing. Higher-order obstructions may still forbid doubly-Artin pairs there, and our results do not assert that such pairs exist.

Accordingly, "base a admits no exclusion class" below always abbreviates "base a admits no *quadratic* exclusion class". Establishing that doubly-Artin pairs actually occur at the remaining gaps is a different and harder problem, of the type addressed under GRH in [14]; we do not address it. The empirical sections confirm that doubly-Artin pairs are abundant outside the exclusion classes, but abundance in a finite range is not an existence theorem.

Since the observation is elementary, we have searched the literature for it specifically before claiming novelty. Moree's hundred-page survey [12] uses the quadratic obstruction $\chi_a(p) = -1$ throughout — it is the source of the entanglement correction in Hooley's density — but always as a condition on a *single* prime; shifts of the argument of χ_a, and the resulting constraints on *pairs* of primes, do not appear. Garcia–Kahoro–Luca [3] and its sequel [4] concern the single gap $g = 2$ and a different statistic — a bias between the twin-prime components in the *number* of primitive roots, $\varphi(p-1)$ versus $\varphi(p+1)$ — not the joint Artin status of the pair; the reversing mechanism plays no role there. Pollack [14] produces, under GRH, bounded-gap runs of primes sharing one fixed primitive root, and Baker–Pollack [1] obtain unconditionally strings of consecutive primes each of which has *some* primitive root drawn from a large prescribed set of primes — the root being allowed to vary with the prime. Neither imposes structure on the gap classes, and neither identifies forbidden ones. We likewise searched for forbidden-gap statements for quadratic characters in other contexts — consecutive quadratic residues and non-residues have a classical literature — and found constraints on character values along progressions, but no statement coupling a gap *class* to the joint primitive-root status of a pair of primes, which is the content here. In that sense "apparently unrecorded" above is meant as the outcome of a search, not as an impression.

Our main theorem determines, for every base, exactly which gap classes are quadratic exclusion classes.

Theorem 1 (Classification of reversing classes; see Theorem 11)**.** *Let a be a non-square positive integer with squarefree part d, let D be the discriminant of $\mathbb{Q}(\sqrt{d})$, and factor $D = D_1D_2\cdots D_k$ into prime discriminants. Then an even shift g is a reversing class for base a if and only if no component D_i is* indeterminate *at g and an odd number of the components* reverse *at g, where the behaviour of each component is given by Table 1.*

Reversal is not the only route to exclusion. Exclusion of doubly-Artin pairs at a gap class requires only that *no admissible residue* r have $\chi_a(r) = \chi_a(r+g) = -1$; reversal is the generic way this happens, but it can also happen because the offending configurations are ruled out by admissibility, i.e. they would force p or $p+g$ to be divisible by an odd prime factor of the conductor. Analysing this second mechanism requires an exact count of the doubly-negative residues, and the engine of the paper is the following identity, which holds for every modulus and requires no primality hypothesis whatever.

Theorem 2 (Counting identity; see Theorem 15)**.** *Let χ be a real character modulo m taking values in $\{0,\pm1\}$, let g be a shift, and set $U(g) = \{r \bmod m : \chi(r) \neq 0,\ \chi(r+g) \neq 0\}$, $T(g) = |U(g)|$, $A(g) = \sum_{r\in U(g)} \chi(r)$, $B(g) = \sum_{r\in U(g)} \chi(r+g)$, $S(g) = \sum_{r\in U(g)} \chi(r)\chi(r+g)$. Then*

$$4\,N_{--}(g) \;=\; T(g) - A(g) - B(g) + S(g), \qquad N_{--}(g) = \#\{\, r \in U(g) : \chi(r) = \chi(r+g) = -1 \,\}.$$

Specialising to prime conductor recovers a closed form, and it is here that a hypothesis is genuinely needed.

Theorem 3 (Inadmissibility exclusion; see Theorem 17)**.** *Let $d \equiv 1 \pmod 4$ be a prime, so that χ_a for $\mathrm{sqf}(a) = d$ has odd conductor d and no reversing classes exist. For $g \not\equiv 0 \pmod d$ the number of residues $r \bmod d$ with $r, r+g \not\equiv 0$ and $\chi(r) = \chi(r+g) = -1$ equals $\bigl(d-3+2\chi(g)\bigr)/4$, while $N_{--}(0) = (d-1)/2$. The count vanishes if and only if $d = 5$ and $\chi(g) = -1$, i.e. $g \equiv 2, 3 \pmod 5$. Consequently base 5 (and any a with $\mathrm{sqf}(a) = 5$) admits the exclusion classes $g \equiv 2, 3 \pmod 5$ — which together contain 40.4% of all consecutive prime pairs — while for prime $d \equiv 1 \pmod 4$, $d \geq 13$, no exclusion class exists.*

The hypothesis $g \not\equiv 0 \pmod d$ is not cosmetic. Since $\chi(0) = 0$, evaluating the displayed formula at $g \equiv 0$ would give $(d-3+2\chi(0))/4 = (d-3)/4$, which is not in general an integer — for $d = 13$ it gives 2.5 against a true count of 6 — so the $g \equiv 0$ case must be handled separately, as it is above. Both branches of the count are machine-checked for $d = 5$ and $d = 13$ (Section 4).

The classification is sharp enough to answer the qualitative question completely.

Corollary 4 (Complete dichotomy; see Corollary 20)**.** *Base a admits at least one quadratic exclusion class if and only if d is even, or $d \equiv 3 \pmod 4$, or $3 \mid d$ (these give reversing classes), or $d = 5$ (inadmissibility classes).*

Thus for $d \in \{13, 17, 29, 37, 41, 53, 61, \dots\}$ there is no quadratic exclusion law at all, while for $d = 3$ fully half of the even gap classes modulo 12 are exclusion classes, and for $d = 5$ two of the five gap classes modulo 5 are excluded. The reversing dichotomy is a genuine structural phenomenon whose source is a degeneration of a Jacobi-sum obstruction at the single prime 3 (Remark 10); the inadmissibility phenomenon is a second, independent degeneration, this time of a counting bound at the single prime 5 (Theorem 17).

In the second half of the paper we turn from the deterministic statement to a statistical one. For a base a define the consecutive-pair correlation

$$\delta(a) \;=\; P\big(p_{n+1} \in \mathrm{A}_a \mid p_n \in \mathrm{A}_a\big) \;-\; P\big(p_{n+1} \in \mathrm{A}_a \mid p_n \notin \mathrm{A}_a\big), \tag{1}$$

computed over consecutive primes $p_n < p_{n+1} \le x$. The bias of Lemke Oliver and Soundararajan [11] in the joint distribution of $(p_n \bmod m, p_{n+1} \bmod m)$ propagates, through the factorization of $p-1$, into Artin status, so one expects $\delta(a) \neq 0$; and indeed we find $\delta(a) < 0$ for all eleven bases examined.

The natural guess is that the exclusion laws drive this anticorrelation: bases whose exclusion classes capture a larger share of consecutive pairs should display a more negative δ. The data complicate this guess in an instructive way. Over all 50,847,531 consecutive prime pairs below 10^9 we find

- the raw association is substantial, $r(w, |\delta|) = 0.646$ over the eleven bases, where w is the weighted fraction of pairs in exclusion classes — but it is confounded with the conductor, since large w occurs only at small f: $r(w, \log f) = -0.643$, and the partial correlation of w with $|\delta|$ given $\log f$ collapses to 0.136;
- the three bases with no exclusion classes at all ($a = 13, 17, 29$) still exhibit $|\delta|$ from 0.022 to 0.043; base 13 alone exceeds seven of the other ten bases (bases $13, 17, 29$ rank fourth, sixth and eighth of eleven), so exclusion structure is not necessary for strong anticorrelation;
- $r(\log f, |\delta|) = -0.957$: conductor size orders the data far better than exclusion weight does;
- restricting δ to gap classes outside the exclusion set (Table 5) gives a *positive* value for all eight bases possessing such classes — a sign reversal under restriction, which is not an additive decomposition of the global value.

What organises the data is the number of residue channels available to χ_a, which is controlled by f: the smaller the conductor, the fewer the classes over which the Lemke Oliver–Soundararajan transition bias is distributed, and the larger the resulting per-base correlation. We state this as a measured empirical regularity (Observation 24) rather than a conjectured exact power of f, because the data do not support a clean exponent.

A note on inferential language. The computation reported here is an exhaustive census of a finite set: every prime below 10^9 is examined, and every count is exact. There is no sampling. We therefore avoid the vocabulary of statistical inference in describing the results. Where a z-score appears in Table 4 it is a nominal, descriptive scale factor computed under a binomial reference model that is certainly false here — consecutive pairs overlap, and the population is deterministic — and it should be read only as the ratio of the effect to that reference scale, not as a test of any hypothesis. The precision of the quoted digits of δ rests on the counts being exact integers, not on any z-value. Correlations across the eleven bases are descriptive summaries of eleven points, not estimates from a random sample of bases.

Summary of results. Section 2 proves the exclusion law and classifies the reversing classes. Section 3 proves the counting identity and derives the inadmissibility mechanism and the complete dichotomy. Section 4 describes the machine verification. Section 5 describes the computation. Section 6 reports the verification of the exclusion law at 10^9. Section 7 reports the measurement of $\delta(a)$, the refutation of the exclusion-density hypothesis, and the conductor regularity. Section 8 discusses what we take the consequences to be.

2. The exclusion law and the reversing classes

Throughout, $a \geq 2$ is an integer that is not a perfect square, $d = \mathrm{sqf}(a)$ denotes its squarefree part, and

$$D = \begin{cases} d, & d \equiv 1 \pmod 4, \\ 4d, & \text{otherwise,} \end{cases} \qquad f = |D|, \tag{2}$$

so that D is the discriminant of $K = \mathbb{Q}(\sqrt{a}) = \mathbb{Q}(\sqrt{d})$ and f is the conductor of the associated quadratic character $\chi_a = \left(\frac{D}{\cdot}\right)$. By quadratic reciprocity χ_a is a Dirichlet character modulo f; in particular

$$\chi_a(p) \text{ depends only on } p \bmod f \qquad (p \nmid 2d). \tag{3}$$

We recall the elementary obstruction, which is the only number-theoretic input to Theorem 8.

Lemma 5. *Let $p > 2$ be a prime with $p \nmid a$. If a is a primitive root modulo p then $\chi_a(p) = -1$.*

Proof. If $\chi_a(p) = +1$ then $a \equiv b^2 \pmod p$ for some b, whence $a^{(p-1)/2} \equiv b^{p-1} \equiv 1 \pmod p$. Since $p > 2$, $(p-1)/2 < p-1$, so the multiplicative order of a is a proper divisor of $p-1$ and a is not a primitive root. □

The converse fails, and this is the reason for the scope restriction announced in Section 1: $\chi_a(p) = -1$ does not make a a primitive root. For instance $\chi_2(43) = -1$, yet 2 has order 14 modulo 43 rather than the full order 42.

Definition 6. Let $R_f = (\mathbb{Z}/f\mathbb{Z})^\times$ be the group of reduced residues modulo f. (By Dirichlet's theorem on primes in arithmetic progressions every class in R_f contains infinitely many primes, so nothing is lost by working with all reduced residues rather than only those realised by primes.) For an even integer g, say that g is

- *reversing* for a if $\chi_a(r+g) = -\chi_a(r)$ for all $r \in R_f$ with $r + g \in R_f$;
- *preserving* for a if $\chi_a(r+g) = +\chi_a(r)$ for all such r;
- *indeterminate* otherwise.

Definition 7. A class $g \bmod f$ is a *quadratic exclusion class* for base a if it admits an even representative — equivalently, any class when f is odd, and an even class when f is even — and there is no residue $r \in R_f$ with $r + g \in R_f$ and $\chi_a(r) = \chi_a(r+g) = -1$. The parity proviso excludes a trivial case: when f is even, an odd g admits no r with r and $r+g$ both odd, so the defining condition would hold vacuously. Since all gaps between odd primes are even, nothing of interest is lost. Every reversing class is an exclusion class (Theorem 8); Theorem 17 below exhibits exclusion classes that are not reversing. As stressed in Section 1, this is a condition on the quadratic character only.

Theorem 8 (Quadratic exclusion law)**.** *Let g be reversing for a. Then there is no pair of primes $p, q > 2$ with $p, q \nmid a$, $q - p \equiv g \pmod f$, such that a is a primitive root modulo both p and q.*

TABLE 1. Behaviour of each prime-discriminant component of χ_a under the shift $r \mapsto r+g$. Here $q \geq 5$ is an odd prime.

component D_i	conductor	reversing iff	preserving iff
-4	4	$g \equiv 2 \pmod 4$	$g \equiv 0 \pmod 4$
8 or -8	8	$g \equiv 4 \pmod 8$	$g \equiv 0 \pmod 8$
-3	3	$g \not\equiv 0 \pmod 3$	$g \equiv 0 \pmod 3$
q^*, $q \geq 5$	q	never	$g \equiv 0 \pmod q$

Proof. By (3) and Definition 6, $\chi_a(q) = \chi_a(p + (q-p)) = -\chi_a(p)$. Hence exactly one of $\chi_a(p), \chi_a(q)$ equals $+1$, and by Lemma 5 the corresponding prime is not Artin base a. $\square$

Theorem 8 is unconditional and its proof is three lines; the content lies in determining the reversing classes, which we now do. Recall that every discriminant D factors uniquely as a product of *prime discriminants*

$$D = D_1 D_2 \cdots D_k, \qquad D_i \in \{-4\} \cup \{8, -8\} \cup \{\, q^* : q \text{ odd prime}\,\}, \tag{4}$$

where $q^* = q$ if $q \equiv 1 \pmod 4$ and $q^* = -q$ if $q \equiv 3 \pmod 4$; the odd D_i are precisely the q^* for the odd primes $q \mid d$, and a factor $\pm 4, \pm 8$ occurs according to $d \bmod 8$. Correspondingly $\chi_a = \prod_i \chi_{D_i}$, with χ_{D_i} of conductor $|D_i|$.

Proposition 9. *Table 1 is correct: each component behaves as stated, and in the cases not listed as reversing or preserving it is indeterminate. In particular a component of conductor 8 is indeterminate for $g \equiv 2, 6 \pmod 8$, and a component q^* with $q \geq 5$ is indeterminate for every $g \not\equiv 0 \pmod q$.*

Proof. Conductor 4. $\chi_{-4}(r) = (-1)^{(r-1)/2}$ depends only on $r \bmod 4$, and on $\{1, 3\}$ it is injective. A shift by $g \equiv 2 \pmod 4$ interchanges 1 and 3, hence reverses; $g \equiv 0 \pmod 4$ fixes both.

Conductor 8. $\chi_8(r) = +1$ iff $r \equiv \pm 1 \pmod 8$, so the fibres are $\{1, 7\}$ and $\{3, 5\}$; the map $r \mapsto r+4$ interchanges these two sets, hence reverses, while $r \mapsto r$ preserves. For $g \equiv 2 \pmod 8$ we have $1 \mapsto 3$ (a reversal) but $3 \mapsto 5$ (a preservation), so the behaviour depends on the class and g is indeterminate; similarly for $g \equiv 6$. The same computation applies to χ_{-8}, whose $+1$-fibre is $\{1, 3\}$ and whose -1-fibre is $\{5, 7\}$, again interchanged by $+4$.

Conductor 3. $\chi_{-3}(r) = +1$ iff $r \equiv 1 \pmod 3$. On the two admissible classes $\{1, 2\}$ the character is injective, so any $g \not\equiv 0 \pmod 3$ moves $1 \leftrightarrow 2$ and reverses, while $g \equiv 0$ preserves. Note that for a fixed such g only one class r satisfies $r, r+g \not\equiv 0 \pmod 3$, so the uniformity required by Definition 6 is a condition on a single residue.

Conductor $q \geq 5$. Suppose, for a contradiction, that $\chi_{q^*}(r+g) = \varepsilon\, \chi_{q^*}(r)$ for a fixed $\varepsilon \in \{\pm 1\}$, all r with $r, r+g \not\equiv 0 \pmod q$, and some $g \not\equiv 0 \pmod q$. Multiplying by $\chi_{q^*}(r)$ and summing over $r \bmod q$ gives

$$S(g) \;:=\; \sum_{r \bmod q} \chi_{q^*}(r)\, \chi_{q^*}(r+g) \;=\; \varepsilon \cdot \#\{\, r : r, r+g \not\equiv 0 \,\} \;=\; \varepsilon\,(q-2). \tag{5}$$

On the other hand, for $g \not\equiv 0 \pmod q$ the substitution $r = gt$ (legitimate since g is invertible) and the complete multiplicativity of χ_{q^*} give

$$S(g) = \sum_{t \bmod q} \chi_{q^*}(gt) \chi_{q^*}(g(t+1)) = \chi_{q^*}(g)^2 \sum_t \chi_{q^*}(t) \chi_{q^*}(t+1) = \sum_{t \bmod q} \chi_{q^*}\left(\frac{t+1}{t}\right),$$

the last sum being over $t \not\equiv 0$. As t runs over the nonzero classes, $u = (t+1)/t = 1 + t^{-1}$ runs over all classes except $u = 1$, whence

$$S(g) = \sum_{u \neq 1} \chi_{q^*}(u) = \left(\sum_{u \bmod q} \chi_{q^*}(u) \right) - \chi_{q^*}(1) = 0 - 1 = -1.$$

Comparing with (5) forces $|-1| = q - 2$, i.e. $q = 3$. For $q \geq 5$ this is a contradiction, so no such ε exists and g is indeterminate. Finally $g \equiv 0 \pmod{q}$ clearly preserves. □

Remark 10. The proof isolates precisely why the prime 3 behaves differently from all larger primes, and it is worth emphasising that this is structural rather than an artifact. The Jacobi-type sum $S(g)$ equals -1 for *every* q and every $g \not\equiv 0$, including $q = 3$; what fails at $q = 3$ is the *counting* side of (5), since $q - 2 = 1$ and $\varepsilon(q-2) = \pm 1$ is then compatible with $S(g) = -1$. Concretely, only one residue class survives the requirement that r and $r + g$ both be prime to 3, and a sign condition imposed on a single class is vacuously uniform. Thus the reversing behaviour of the component -3 — which is what makes base 3 the most heavily constrained base — sits exactly at the boundary where the obstruction of Proposition 9 degenerates.

Theorem 11 (Classification of reversing and preserving classes)**.** *Let a be non-square with $D = D_1 \cdots D_k$ as in (4), and let g be even. Then g is reversing for a if and only if no D_i is indeterminate at g and the number of i with D_i reversing at g is odd; and g is preserving if and only if no D_i is indeterminate at g and that number is even.*

Proof. Since $\chi_a = \prod_i \chi_{D_i}$ and the moduli $|D_i|$ are pairwise coprime, the Chinese Remainder Theorem identifies R_f with the product of the component residue sets. If every component is reversing or preserving at g, then for every r, $\chi_a(r+g) = \prod_i \chi_{D_i}(r+g) = \left(\prod_i \varepsilon_i\right)\chi_a(r)$ where $\varepsilon_i = -1$ exactly for the reversing components; so χ_a is reversed iff the count of such i is odd. Conversely, suppose some component D_j is indeterminate at g, so there are r_j, r'_j in its residue set with $\chi_{D_j}(r_j + g) = -\chi_{D_j}(r_j)$ and $\chi_{D_j}(r'_j + g) = +\chi_{D_j}(r'_j)$. Fixing any admissible choices at the other components and varying only the j-th coordinate via CRT produces two residues $r, r' \in R_f$ at which χ_a is respectively reversed and preserved. Hence g is neither reversing nor preserving. □

Theorem 11 classifies the reversing and preserving classes; it does not by itself classify the exclusion classes of Definition 7, since an indeterminate class may still be an exclusion class. That gap is closed in Section 3.

Corollary 12 (Dichotomy for reversing classes)**.** *Base a admits at least one* reversing *class if and only if*

$$d \equiv 0 \pmod{2}, \qquad \text{or} \qquad d \equiv 3 \pmod{4}, \qquad \text{or} \qquad 3 \mid d,$$

equivalently if and only if $\{-4, 8, -8, -3\} \cap \{D_1, \ldots, D_k\} \neq \emptyset$.

Proof. By Theorem 11 a reversing g requires at least one component that is reversing at g, and by Table 1 the only components capable of reversing are -4, ± 8 and -3. The component -4 occurs iff $d \equiv 3 \pmod{4}$, a component ± 8 occurs iff d is even, and -3 occurs iff $3 \mid d$; this gives the stated equivalence of conditions.

Conversely, suppose such a component D_j exists. Choose g by CRT to lie in a reversing class of D_j and in the class 0 modulo $|D_i|$ for every $i \neq j$; this is possible as the moduli are coprime. Then every other component preserves and exactly one reverses, so g is reversing by Theorem 11. It remains to check that g may be taken even. If f is even, then D has a 2-primary prime discriminant component -4 or ± 8, and the CRT prescription above assigns g

the residue 0 or 2 modulo 4, or 0 or 4 modulo 8; in every case the prescribed residue is even, so every CRT solution g is even. (Note that adding f would not help here, since f even preserves parity.) If f is odd then $D_j = -3$ and $f = |D| = d$ is odd; here the reversing condition is $g \not\equiv 0 \pmod 3$ together with $g \equiv 0$ modulo the other odd components, and since $\gcd(2, f) = 1$ every class modulo f contains even integers, so again an even representative exists. $\square$

Remark 13. The parity step above deserves comment, because a natural shortcut is false. One might argue that f is even unless no reversing component exists; that is incorrect. The base $a = 21$ has $d = 21 \equiv 1 \pmod 4$, hence odd conductor $f = 21$, yet it has the reversing component -3 and genuinely possesses reversing classes, namely $g \equiv 7, 14 \pmod{21}$ (see Table 2). Because f is odd, each of these classes contains even integers — for example $g = 28$ and $g = 14$ — so the exclusion law applies to actual even prime gaps. The correct statement is the one used in the proof: when f is odd, every residue class modulo f contains even integers.

Remark 14. The bases that admit no reversing class are exactly those whose squarefree part satisfies $d \equiv 1 \pmod 4$ with d odd and $3 \nmid d$, that is, $d \in \{5, 13, 17, 29, 37, 41, 53, 61, \dots\}$. Among $2 \le a \le 30$ these are $a = 5, 13, 17, 20, 29$ (note $\mathrm{sqf}(20) = 5$). At the other extreme, the case $d = 3$ gives $f = 12$, and *three* of the six even classes modulo 12 are reversing. This is special to $d = 3$: other $d \equiv 3 \pmod 4$ with $3 \mid d$ have larger conductors, for instance $f = 60$ when $d = 15$, and $f = 156$ when $d = 39$. Bases a with $\mathrm{sqf}(a) = 3$, such as $a = 3$, 12, 27 and 48, all share the conductor $f = 12$, and hence the same three reversing classes.

3. The counting identity and the complete dichotomy

For the reversal-free bases the conductor $f = d$ is odd, and one might expect — as an earlier version of this paper asserted — that no exclusion of any kind is possible. That is wrong, for exactly one squarefree class, and the data flagged it before we understood it: at $x = 10^9$ base 5 has *zero* doubly-Artin consecutive pairs at every gap $g \equiv 2, 3 \pmod 5$, across 20,523,554 pairs. The explanation is not reversal but a counting degeneration, and to analyse it we need an exact count of the doubly-negative residues.

We isolate that count as a standalone identity. It requires no primality assumption and no character theory: it is a two-line inclusion–exclusion, stated in the generality in which we shall use it.

Theorem 15 (Counting identity)**.** *Let U be a finite set and let $\varphi, \psi : U \to \{-1, +1\}$. Put*

$$N = \#\{r \in U : \varphi(r) = \psi(r) = -1\}, \qquad T = |U|,$$

$$A = \sum_{r \in U} \varphi(r), \qquad B = \sum_{r \in U} \psi(r), \qquad S = \sum_{r \in U} \varphi(r)\psi(r).$$

Then $4N = T - A - B + S$.

Proof. For $x, y \in \{-1, +1\}$ one checks the four cases directly: $(1-x)(1-y)$ equals 4 if $x = y = -1$ and equals 0 otherwise. Hence $4 \cdot \mathbf{1}[\varphi(r) = \psi(r) = -1] = (1 - \varphi(r))(1 - \psi(r))$ for every $r \in U$. Summing over U and expanding the product gives $4N = \sum_{r \in U} \big(1 - \varphi(r) - \psi(r) + \varphi(r)\psi(r)\big) = T - A - B + S$. $\square$

We apply this with $U = U(g) := \{r \bmod m : \chi(r) \ne 0,\ \chi(r + g) \ne 0\}$ and $\varphi(r) = \chi(r)$, $\psi(r) = \chi(r + g)$, for a real character χ modulo m; both take values in $\{\pm 1\}$ on $U(g)$ by construction. Writing $T(g), A(g), B(g), S(g), N_{--}(g)$ for the resulting quantities, Theorem 15 reads

$$4\, N_{--}(g) \;=\; T(g) - A(g) - B(g) + S(g). \tag{6}$$

Note that $S(g)$ may equally be summed over all residues modulo m, since the terms outside $U(g)$ vanish.

Everything in this section is a consequence of (6) together with an evaluation of the four quantities. The advantage over an argument conducted at the level of sign patterns is that T, A, B, S are all multiplicative under the Chinese Remainder Theorem, whereas the property "some residue realises the pattern $(-1,-1)$" is not usefully so; this is what makes the composite case of Corollary 20 tractable.

Lemma 16 (Component evaluation)**.** *Let $q \geq 5$ be prime and let $\chi_q = \left(\frac{\cdot}{q}\right)$ be the Legendre symbol. For a shift g:*

$$(T_q, A_q, B_q, S_q) = \begin{cases} (q-1,\ 0,\ 0,\ q-1), & q \mid g, \\ (q-2,\ -\chi_q(-g),\ -\chi_q(g),\ -1), & q \nmid g. \end{cases}$$

Proof. If $q \mid g$ then $\chi_q(r+g) = \chi_q(r)$, so U_q is the set of $q-1$ nonzero residues, $A_q = B_q = \sum_{r\neq 0} \chi_q(r) = 0$, and $S_q = \sum_{r\neq 0} \chi_q(r)^2 = q-1$.

If $q \nmid g$ then U_q omits exactly the two distinct residues $r \equiv 0$ and $r \equiv -g$, so $T_q = q-2$. For A_q we have $\sum_{r \bmod q} \chi_q(r) = 0$, and the omitted terms contribute $\chi_q(0) = 0$ and $\chi_q(-g)$, whence $A_q = -\chi_q(-g)$; symmetrically, substituting $r \mapsto r-g$ gives $B_q = -\chi_q(g)$. Finally $S_q = -1$ by the Jacobi-type evaluation established in the proof of Proposition 9. □

Theorem 17 (Inadmissibility exclusion)**.** *Let $d \equiv 1 \pmod 4$ be prime and $\chi = \left(\frac{\cdot}{d}\right)$. Then*

$$N_{--}(g) = \begin{cases} \dfrac{d-1}{2}, & g \equiv 0 \pmod d, \\ \dfrac{d-3+2\chi(g)}{4}, & g \not\equiv 0 \pmod d. \end{cases}$$

In particular $N_{--}(g) = 0$ if and only if $d = 5$ and $\chi(g) = -1$, i.e. $g \equiv 2, 3 \pmod 5$. For every prime $d \equiv 1 \pmod 4$ with $d \geq 13$, and every g, we have $N_{--}(g) \geq 2$, so no exclusion class exists.

Proof. Apply (6) with the values of Lemma 16 (whose proof used only $q \geq 5$ prime, so it applies to d).

If $d \mid g$ then $4N_{--} = (d-1) - 0 - 0 + (d-1) = 2(d-1)$, giving $N_{--} = (d-1)/2$.

If $d \nmid g$ then, since $d \equiv 1 \pmod 4$ gives $\chi(-1) = +1$ and hence $\chi(-g) = \chi(g)$,

$$4N_{--} = (d-2) + \chi(g) + \chi(g) - 1 = d - 3 + 2\chi(g),$$

as claimed. This vanishes iff $d - 3 = -2\chi(g)$, i.e. $\chi(g) = -1$ and $d = 5$; and for $d \geq 13$ it is at least $(13-3-2)/4 = 2$. The case $d \mid g$ never vanishes since $(d-1)/2 \geq 2$ for $d \geq 5$. □

Remark 18. The distinction between the two cases is essential and was omitted in an earlier version of this paper. Applying the second formula at $g \equiv 0$ would give $(d-3)/4$, which for $d = 13$ is 2.5 — not an integer, against a true count of 6. At $d = 5$ the incorrect extension returns $1/2$ against a true count of 2, so the failure is equally visible there; we do not know why it survived. We record it because the same class of error — a character-sum evaluation applied outside the range where its input identity holds — is easy to make and hard to see.

Remark 19. The exact count explains the phenomenon structurally: the constraint "both characters -1" costs a factor of roughly 4 in density, and the two inadmissible residues $r \equiv 0, -g$ cost $\chi(g)$-dependent boundary terms; only at $d = 5$, where just $(d-1)/2 = 2$ residues carry $\chi = -1$ at all, can the boundary terms consume the entire count. It is a degeneration at small conductor, exactly parallel to — but independent of — the $q = 3$ degeneration of Remark 10.

We can now settle the composite case, which the component-evaluation lemma makes routine.

Corollary 20 (Complete dichotomy)**.** *Base a with squarefree part d admits at least one quadratic exclusion class (Definition 7) if and only if*

$$d \equiv 0 \pmod 2, \qquad d \equiv 3 \pmod 4, \qquad 3 \mid d, \qquad \textit{or} \qquad d = 5.$$

Proof. The first three cases give reversing classes (Corollary 12), which are exclusion classes by Theorem 8, and $d = 5$ gives the inadmissibility classes of Theorem 17.

Conversely, suppose $d \equiv 1 \pmod 4$, d odd, $3 \nmid d$, $d \neq 5$; we must show $N_{--}(g) > 0$ for every g. Then $f = d$ and $\chi = \prod_{q|d} \chi_q$, the product over the k distinct prime factors of d, each satisfying $q \geq 7$ or $q = 5$; in all cases $q \geq 5$. The case $k = 1$ is Theorem 17, so assume $k \geq 2$. By CRT the quantities T, A, B, S factor as products of their components, $T(g) = \prod_q T_q(g)$ and likewise for A, B, S, with the values of Lemma 16.

Case 1: $q_0 \mid g$ for some $q_0 \mid d$. Then $A_{q_0} = B_{q_0} = 0$, so $A(g) = B(g) = 0$ and (6) gives $4N_{--}(g) = T(g) + S(g)$. Now $T_q \geq |S_q|$ for every q, with equality only when $q \mid g$ (where both equal $q - 1$); and when $q \nmid g$ we have $T_q = q - 2 \geq 3 > 1 = |S_q|$. Hence $T(g) \geq |S(g)|$, with strict inequality unless $q \mid g$ for every $q \mid d$. If the inequality is strict then $4N_{--}(g) > 0$; if not, then $d \mid g$ and $4N_{--}(g) = 2\prod_q(q - 1) > 0$. Either way $N_{--}(g) > 0$.

Case 2: $\gcd(g, d) = 1$. Then every component is in the second case of Lemma 16, so $T(g) = \prod_q(q - 2)$, $A(g) = (-1)^k\chi(-g)$, $B(g) = (-1)^k\chi(g)$ and $S(g) = (-1)^k$. Since $d \equiv 1 \pmod 4$ we have $\chi(-1) = +1$, so $\chi(-g) = \chi(g)$ and

$$4N_{--}(g) = \prod_{q|d}(q - 2) \; - \; (-1)^k\big(2\chi(g) - 1\big).$$

The bracket lies in $\{1, -3\}$, so $4N_{--}(g) \geq \prod_q(q - 2) - 3$. Every $q \mid d$ satisfies $q \geq 5$, so $q - 2 \geq 3$ and $\prod_q(q - 2) \geq 3^k \geq 9$ for $k \geq 2$. Hence $4N_{--}(g) \geq 6 > 0$.

In both cases $N_{--}(g) > 0$, so no exclusion class exists. $\square$

Remark 21. The composite case above replaces an argument in an earlier version of this paper which attempted to construct the pattern $(-1, -1)$ component by component and then combine the choices by CRT. That argument is not valid: it requires each component to admit mixed sign patterns, which fails exactly when $q \mid g$. For $d = 65$ and $g = 10$, the component $q = 5$ admits only the patterns $(+, +)$ and $(-, -)$, since $\chi_5(r + 10) = \chi_5(r)$ identically, so no component-wise choice of a mixed pattern is available there. Working with the aggregate quantities T, A, B, S — which are multiplicative — avoids the difficulty entirely, and the case $q \mid g$ becomes the *easier* of the two.

Remark 22. The bound in Case 2 is close to sharp. For $d = 385 = 5 \cdot 7 \cdot 11$ one has $\prod(q - 2) = 135$ and the minimum of $4N_{--}(g)$ over g coprime to d is $132 = 135 - 3$, attained where $\chi(g) = -1$ and k is odd.

Computational confirmation of the classification. For each non-square a with $2 \leq a \leq 30$ we computed the reversing and preserving sets directly, by evaluating χ_a on every reduced residue class modulo f and testing Definition 6 exhaustively over every gap class admitting an even representative — all g modulo f when f is odd, and the even g when f is even. The resulting sets agree *exactly*, in every case, with the prediction of Theorem 11. The exact count of Theorem 17 was verified for all primes $d \equiv 1 \pmod 4$ with $d \leq 41$ and all shifts, in both the $g \equiv 0$ and $g \not\equiv 0$ cases. The identity (6) was verified directly against brute-force counts for every non-square $a \leq 120$ and every shift modulo f, a total of 14,803 (base, shift) pairs, with no discrepancy. The criterion of the complete dichotomy (Corollary 20) was confirmed by exhaustive scan for all 72 non-square a with $2 \le a \le 80$, again with no discrepancy; for odd conductors this scan covers every residue class, odd residues included, and for even conductors the even classes, in accordance with the parity proviso of Definition 7. The identity check of (6), by contrast, was run over *all* shifts modulo f without that filter. Table 2 records a representative sample.

4. Machine verification

Part of the deterministic half of this paper has been formalised and checked by the Lean 4 proof assistant against `mathlib`: the counting identity in full generality, and the prime-conductor count in the two special cases $d = 5$ and $d = 13$. We report this because both statements were, in an earlier version, asserted in forms that were false or incomplete, and a proof assistant is an effective guard against exactly that failure.

The formalisation covers:

- the counting identity of Theorem 15, in the general form stated there — an arbitrary finite set and arbitrary $\{\pm 1\}$-valued functions, with no modulus and no primality;
- the two-case count of Theorem 17 for the conductors $d = 5$ and $d = 13$, including an explicit statement that the $g \not\equiv 0$ formula *fails* at $g \equiv 0$ when $d = 13$ (there $4N_{--} = 24$ while the formula returns 10);
- the vanishing pattern: for $d = 5$ the count vanishes exactly on $g \equiv 2, 3 \pmod 5$, and for $d = 13$ it never vanishes.

The Lean development contains no `sorry`, no additional axioms and no appeal to `native_decide`; every theorem depends only on Lean's three standard axioms (`propext`, `Classical.choice`, `Quot.sound`), and a check script verifies this. We state plainly what is *not* formalised. The general-d case of Theorem 17 is not, because it requires the Jacobi-type evaluation $\sum_r \chi(r)\chi(r+g) = -1$. The underlying ingredient is in fact present in `mathlib` as `jacobiSum_nontrivial_inv` ($J(\chi, \chi^{-1}) = -\chi(-1)$, which for quadratic χ gives the shifted sum after the substitution $r = -gx$); what this development has not done is formalise the specialisation and the bridge to an integer-valued shifted sum. The reversing classification of Theorem 11 is not formalised either, and no empirical statement in Sections 5–7 is formalised. The formalisation is a check on the two most error-prone statements, not a certificate for the paper.

5. Data and computation

We enumerated the primes up to $x = 10^9$ by a segmented sieve of Eratosthenes with segment length 2^{22}, obtaining 50,847,532 primes $p \ge 5$ and hence 50,847,531 consecutive pairs. For each prime p we factored $p - 1$ once, by trial division over the primes below 10^5 together with the cofactor, and then tested primitive-root status for all of the bases

$$a \in \{2, 3, 5, 6, 7, 10, 11, 13, 17, 21, 29\}$$

simultaneously, using the standard criterion that, for $p \nmid a$, a is a primitive root modulo p iff $a^{(p-1)/\ell} \not\equiv 1 \pmod p$ for every prime $\ell \mid p - 1$. (The hypothesis $p \nmid a$ is necessary: for $a = p$ the displayed condition holds vacuously while a is not a unit. The implementation tests $p \nmid a$ first.) Modular exponentiation was carried out in 128-bit arithmetic. Sharing the factorization of $p - 1$ across all eleven bases is what makes the computation inexpensive; the run completed in under an hour on a single core of the author's machine, though we did not record a timed benchmark and writes no intermediate table, accumulating instead the 2×2 contingency counts of (Artin status of p_n, Artin status of p_{n+1}), both globally and separately for each gap $g \le 300$.

TABLE 2. Exclusion classes for selected bases. "Prime discs." lists the factorization (4). Reversing classes are those predicted by Theorem 11; inadmissibility classes by Theorem 17; all confirmed by exhaustive search. For odd conductors a residue may be odd even though the gaps representing it are even — every class then admits an even representative, e.g. $g \equiv 7 \pmod{21}$ via $g = 28, 70, \dots$ — while other residues, such as 2 mod 5 and 14 mod 21, are already even. An earlier version of this table, which scanned only even residues, missed the class 7 of base 21 and all classes of base 5.

a	f	prime discs.	reversing	inadmissibility	preserving
2	8	8	4	—	0
3	12	$-3, -4$	$4, 6, 8$	—	$0, 2, 10$
5	5	5	*none*	$2, 3$	0
6	24	$-3, -8$	$8, 12, 16$	—	$0, 4, 20$
7	28	$-7, -4$	14	—	0
10	40	$5, 8$	20	—	0
11	44	$-11, -4$	22	—	0
13	13	13	*none*	*none*	0
15	60	$-3, 5, -4$	$20, 30, 40$	—	$0, 10, 50$
17	17	17	*none*	*none*	0
21	21	$-3, -7$	$7, 14$	—	0
29	29	29	*none*	*none*	0
30	120	$-3, 5, -8$	$40, 60, 80$	—	$0, 20, 100$

The base set was chosen to span the dichotomy as we then understood it. In the corrected classification (Corollary 20): $a = 13, 17, 29$ admit no exclusion law; $a = 3$ and $a = 6$ have three reversing classes each; $a = 2, 7, 10, 11$ have one reversing class; $a = 21$ has two; and $a = 5$ has the two inadmissibility classes of Theorem 17. All counts reported below are exact; no sampling is involved. The implementation, together with an independent and slower verification in `sympy` used to cross-check the exclusion counts at 3×10^6, is available at the repository cited in the data availability statement.

6. Verification of the exclusion law at 10^9

Theorem 8 is a proved statement, so a computation cannot strengthen it; but it is a sharp and easily falsified prediction about a large finite set, and checking it guards against errors in the classification of Theorem 11 and in the implementation. For each base with at least one exclusion class we counted the consecutive prime pairs (p_n, p_{n+1}) whose gap lies in an exclusion class, and among those the pairs that are doubly Artin. Table 3 reports the result.

The eight bases with exclusion classes comprise seven with reversing classes and one — base 5 — whose classes arise from inadmissibility. Summing over bases gives 84,981,870 base–pair incidences, and not a single doubly-Artin pair among them. Note that this total counts incidences, not distinct prime pairs: there are only 50,847,531 consecutive pairs in all, and a pair whose gap lies in exclusion classes of several bases contributes once per base. Among the incidences is the class $g \equiv 7 \pmod{21}$ of base 21, which is odd as a residue but is realised by even gaps $g = 28, 70, \dots$ since the conductor 21 is odd.

At the preserving classes $g \equiv 0 \pmod f$ the doubly-Artin proportion lies between 26.3% and 29.8% for the eight bases of Table 3. (Across all eleven bases the range is wider: base

TABLE 3. Exclusion law at $x = 10^9$. "Pairs" counts consecutive prime pairs whose gap lies in an exclusion class for the given base; "both Artin" counts those in which a is a primitive root modulo both primes. The control column gives, for comparison, the proportion of doubly-Artin pairs at the preserving class $g \equiv 0$. The total is a sum of base–pair incidences: a single prime pair is counted once for each base whose exclusion classes contain its gap.

a	f	exclusion classes	pairs	both Artin	control (preserving)
2	8	4 (rev.)	13,404,106	0	27.5%
3	12	$4, 6, 8$ (rev.)	27,010,759	0	28.8%
5	5	$2, 3$ (inadm.)	20,523,554	0	29.8%
6	24	$8, 12, 16$ (rev.)	12,419,039	0	29.3%
7	28	14 (rev.)	3,567,335	0	26.3%
10	40	20 (rev.)	2,214,511	0	26.7%
11	44	22 (rev.)	1,796,191	0	27.5%
21	21	$7, 14$ (rev.)	4,046,375	0	27.5%
total incidences			84,981,870	**0**	

29, which has no exclusion classes and so does not appear in the table, has a preserving-class proportion of 24.9%.) We do not offer a closed-form explanation of that range: the natural guess, that it should be the square of the relevant Artin density, does not fit, since Artin's constant gives $C^2 = 0.13984$ and the base-5 variant gives $(20C/19)^2 = 0.15495$, both far below the observed values. The discrepancy is unsurprising — conditioning on $g \equiv 0 \pmod{f}$ is a strong constraint on $p \bmod f$ and the two Artin statuses in a pair are not independent — but we have not quantified it, and the column should be read purely as an empirical control showing that doubly-Artin pairs are common at preserving gaps.

To calibrate how striking the zero count is: treating Artin statuses as independent with the observed per-base densities, the expected number of doubly-Artin incidences in the exclusion classes would be about 12,189,000 — some 3.8 million for base 3 alone and 3.2 million for base 5. The observed count is 0. This is a calibration of the *verification*, not of the theorems, which make the zero certain; it indicates the scale of discrepancy the check was capable of detecting, though it does not by itself bound the size of an error that could have escaped notice. A supplementary check with independent `sympy` order computations for primes below 3×10^6 reproduced zero doubly-Artin counts, but it covers a strict subset of Table 3: bases $2, 3, 6, 7, 10, 11$ and 21, and for base 21 only the class 14. It does not cover base 5, nor the class 7 of base 21. The zero counts for those cases rest on the primary computation alone.

7. THE CONSECUTIVE-PAIR CORRELATION ACROSS BASES

We now measure $\delta(a)$ of (1). Write n_{ij} for the number of consecutive pairs with Artin status i at p_n and j at p_{n+1}, so that

$$\widehat{\delta}(a) = \frac{n_{11}}{n_{10} + n_{11}} - \frac{n_{01}}{n_{00} + n_{01}}.$$

Table 4 collects the results.

Two features stand out. First, $\delta(a) < 0$ for every base: consecutive primes *repel* in Artin status, uniformly in the base. This is consistent with the mechanism proposed for the base-10 case, namely that the Lemke Oliver–Soundararajan bias [11] in the joint distribution of

TABLE 4. Consecutive-pair Artin correlation at $x = 10^9$, over all 50,847,531 consecutive prime pairs, sorted by $|\delta|$. Here #exc is the number of exclusion classes (complete classification, reversing and inadmissibility) and w the fraction of all consecutive pairs whose gap lies in one. The column z is a nominal binomial scale factor, not a hypothesis test; see the note on inferential language in Section 1.

a	f	#exc	w	$\widehat{\delta}(a)$	z (nominal)
5	5	2	0.4036	−0.066563	−478.9
2	8	1	0.2636	−0.050199	−361.0
3	12	3	0.5312	−0.046676	−335.4
13	13	0	0.0000	−0.042573	−305.6
21	21	2	0.0796	−0.038345	−275.2
17	17	0	0.0000	−0.033025	−236.7
6	24	3	0.2442	−0.025205	−180.4
29	29	0	0.0000	−0.022422	−160.4
11	44	1	0.0353	−0.018909	−135.2
10	40	1	0.0436	−0.014140	−101.0
7	28	1	0.0702	−0.013491	−96.4

consecutive prime residues propagates through the factorization of $p-1$ — and hence through $\omega(p-1)$, whose consecutive values are themselves anticorrelated — into Artin status.

Second, and contrary to what the first half of this paper might suggest, the ordering of $|\delta(a)|$ is not explained by the exclusion law.

Observation 23 (Conductor outperforms exclusion weight as a descriptive predictor)**.** With the complete classification (including the inadmissibility classes of base 5), the exclusion weight w and $|\delta|$ correlate at $r(w, |\delta|) = 0.646$ at $x = 10^9$. That is not negligible; but after linear adjustment for $\log f$ little of it remains, and the ordering of $|\delta|$ does not track w:

(1) it is confounded with the conductor, $r(w, \log f) = -0.643$, and the partial correlation of w with $|\delta|$ given $\log f$ is 0.136;
(2) the three bases with no exclusion classes at all ($a = 13, 17, 29$) carry $|\delta| = 0.043$, 0.033 and 0.022 respectively. Base 13 ranks fourth of the eleven, exceeding seven other bases, so exclusion structure is not necessary for strong anticorrelation;
(3) restricting δ to the pairs *outside* the exclusion classes (Table 5) yields a positive value for *all eight* bases possessing such classes (e.g. base 3: -0.047 globally, $+0.578$ outside; base 5: -0.067 globally, $+0.002$ outside), while the no-exclusion bases are untouched by construction. This is a sign reversal under restriction; as explained after Table 5, it is not an additive decomposition of the global δ, and no contribution of the exclusion classes to the global value can be read off from it.

These are descriptive summaries of eleven points; the eleven bases are not a random sample, and no inference to other bases is intended.

Table 5 should be read with care. It restricts the statistic to a subset of gap classes; it is *not* an additive decomposition of the global δ, and no contribution of the exclusion classes to the global value can be read off from it. The two conditional proportions defining δ carry different gap-class weights in the Artin and non-Artin populations, so restriction and aggregation do not commute. What the table does establish is a sign reversal: for each of the eight bases with

TABLE 5. Global δ versus δ restricted to gap classes outside the exclusion set, at $x = 10^9$. w is the weighted fraction of pairs in exclusion classes (complete classification). All eight bases with exclusion classes have positive δ outside them.

a	w	δ (global)	δ (outside exclusion)
5	0.4036	-0.0666	$+0.0020$
2	0.2636	-0.0502	$+0.1442$
3	0.5312	-0.0467	$+0.5782$
13	0	-0.0426	-0.0426
21	0.0796	-0.0383	$+0.0107$
17	0	-0.0330	-0.0330
6	0.2442	-0.0252	$+0.1581$
29	0	-0.0224	-0.0224
11	0.0353	-0.0189	$+0.0020$
10	0.0436	-0.0141	$+0.0131$
7	0.0702	-0.0135	$+0.0319$

exclusion classes, the statistic computed outside those classes is positive, while the global value is negative.

We note also that the global anticorrelation is not purely a within-gap-class phenomenon even for the bases with no exclusion classes. Writing the global δ against the pair-weighted mean of the per-gap-class values, the difference — a between-class term — is -0.0074 for $a = 13$, -0.0110 for $a = 17$ and -0.0119 for $a = 29$, so a substantial part of the effect lives in the variation of Artin density across gap classes rather than within them. We record these as measured quantities and do not attempt a mechanism.

What does organise the data is the conductor.

Observation 24 (Conductor regularity)**.** Over the eleven bases at $x = 10^9$,

$$r(\log f, |\delta|) = -0.957, \qquad r(f^{-1/2}, |\delta|) = +0.951, \qquad r(f^{-1}, |\delta|) = +0.921,$$

against $r(w, |\delta|) = +0.646$ for the exclusion density, whose partial correlation given $\log f$ is 0.136. Thus $|\delta(a)|$ decreases with the conductor of χ_a, and among the quantities compared here it orders the observed data better than the exclusion weight does. The dependence is not perfectly monotone: $a = 21$ with $f = 21$ has $|\delta| = 0.0383$, exceeding $a = 17$ with the smaller conductor $f = 17$ and $|\delta| = 0.0330$.

The heuristic is that χ_a is determined modulo f, so the residue information relevant to Artin status base a is carried by the $\varphi(f)$ classes in $(\mathbb{Z}/f\mathbb{Z})^\times$. The transition bias of [11] is distributed across these channels; the fewer the channels, the less the cancellation between them, and the larger the surviving global correlation. Figure 1 displays both associations.

We stress that Observation 24 is at present a purely empirical regularity, and we indicate what a derivation would involve. Let X and Y be the Artin indicators of p_n and p_{n+1}. Then $\delta(a) = \mathrm{Cov}(X, Y)/\big(P(X = 1)(1 - P(X = 1))\big)$, so it suffices to model the covariance. Assuming that X and Y are conditionally independent given the residue pair $(p_n, p_{n+1}) \bmod f$ — an approximation, since the factorizations of $p_n - 1$ and $p_{n+1} - 1$ share structure not captured by

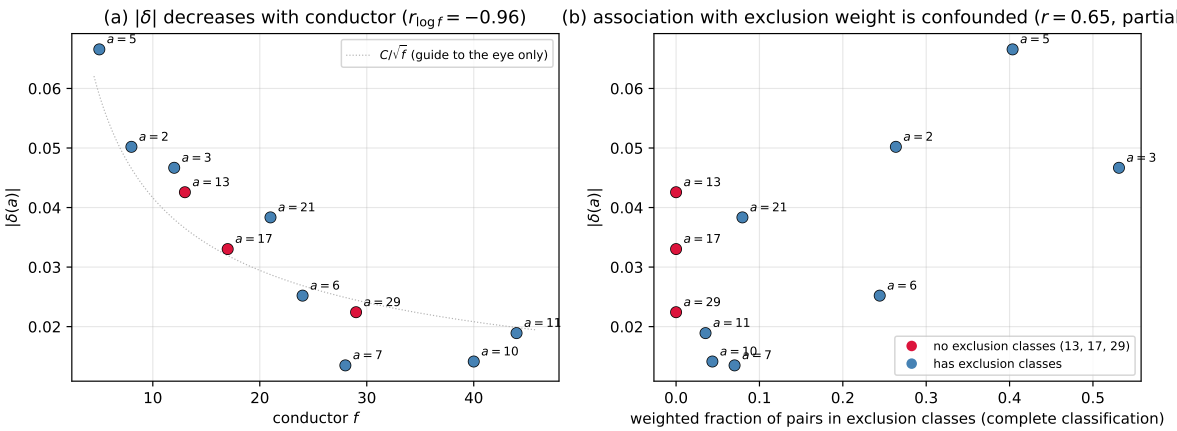

FIGURE 1. Left: $|\delta(a)|$ against the conductor f of χ_a at $x = 10^9$. The data points are the content of the figure; the faint dotted curve $C/\sqrt{f}$ ($C = 0.132$) is a guide to the eye only — as discussed after Observation 24, the data do not identify an exponent and no power law is asserted. Right: $|\delta(a)|$ against the weighted fraction of consecutive pairs lying in exclusion classes (complete classification, including the inadmissibility classes of base 5). The three bases with no exclusion classes ($a = 13, 17, 29$) are shown in red; the apparent association in this panel is confounded with the conductor (see Observation 23).

residues modulo f — one obtains

$$\mathrm{Cov}(X, Y) \;\approx\; \sum_{r,s \,\in\, (\mathbb{Z}/f\mathbb{Z})^\times} \Big(\pi(r,s) - \pi(r)\pi(s)\Big)\, \alpha_a(r)\, \alpha_a(s),$$

where $\pi(r,s)$ is the density of consecutive prime pairs with $p_n \equiv r$, $p_{n+1} \equiv s \pmod f$, and $\alpha_a(r)$ is the deviation of the Artin density in the class r from its mean; a residual term from the failure of conditional independence has been dropped. The first factor is governed, conjecturally, by the Hardy–Littlewood singular series [6] through the conjectural formula of Lemke Oliver–Soundararajan [11, Conj. 1.1] for prime transitions in residue classes; the second by Hooley's conditional density computation [8] restricted to residue classes, in which $\chi_a(r) = -1$ is the leading constraint. Carrying this out — in particular extracting the f-dependence of the double sum, where the number of channels $\varphi(f)^2$ competes against the per-channel bias — would replace the measured correlations above with a formula and settle the exponent question. We have not done this here: the LOS transition term for *consecutive* (rather than merely distinct) primes in fixed residue classes involves a secondary term with no closed form, and the interaction with the $\omega(p-1)$ channel (through which the bias reaches Artin status) is not multiplicative. The calculation appears feasible but is a paper of its own, and we leave it as the first item of future work.

We deliberately refrain from asserting an exact power law. Across the eleven bases the products $|\delta| \cdot f$ range over 0.333 to 0.832, while $|\delta| \cdot \sqrt{f}$ range over 0.0714 to 0.1757; the latter is the narrower band, so $f^{-1/2}$ is the better one-parameter description, but neither product is constant and neither varies monotonically with f. The extremes of the $|\delta|\sqrt{f}$ band are $a = 7$ (0.0714) and $a = 21$ (0.1757), a factor of 2.5 apart, so the fitted constant $C = 0.132$ in Figure 1 summarises a genuinely dispersed set of values. With eleven bases spanning $5 \le f \le 44$ we regard the evidence as insufficient to identify an exponent. The honest statement is Observation 24: a strong but imperfect monotone dependence on f, whose precise shape should be derived from, rather than fitted ahead of, a channel-by-channel analysis.

Remark 25. The two smallest $|\delta|$ belong to $a = 7$ and $a = 10$, with $f = 28$ and $f = 40$; the largest to $a = 5$ with $f = 5$. As noted in Observation 24, $a = 21$ with $f = 21$ sits above both $a = 17$ ($f = 17$) and $a = 6$ ($f = 24$), so the dependence on f is not perfectly monotone — as one would expect if the relevant quantity is a sum over the $\varphi(f)$ channels rather than a function of f alone.

8. Discussion

What is new here. The ingredients are classical: the quadratic obstruction to being a primitive root, quadratic reciprocity, the decomposition of a discriminant into prime discriminants, and the elementary evaluation of a Jacobi-type character sum. The contributions are (i) the observation that these combine into *deterministic quadratic exclusion laws* for consecutive Artin primes (Theorem 8); (ii) the identification of *two distinct mechanisms* producing them — reversal, classified completely in terms of prime discriminants (Theorem 11), and inadmissibility, governed by the exact count of Theorem 17, which degenerates precisely at $d = 5$ — together with the complete dichotomy (Corollary 20), all resting on the counting identity of Theorem 15; and (iii) the first cross-base measurement of the consecutive-pair Artin correlation, in which conductor size orders the eleven measured values better than exclusion weight does.

We call attention to a corrective aspect of (ii). An earlier version of this paper classified only the reversing mechanism and asserted that base 5 admits no exclusion law. The data refuted the assertion — twenty million consecutive pairs at gaps $g \equiv 2, 3 \pmod 5$ with a zero doubly-Artin count is not a subtle signal — and the inadmissibility mechanism was found by asking what the classification had missed. That same earlier version contained two further defects, both since corrected: the count of Theorem 17 was stated without the hypothesis $g \not\equiv 0 \pmod d$, under which it is false and not even integer-valued (Remark 18); and the composite case of the dichotomy was argued by a component-wise construction that fails when a prime factor of d divides the shift (Remark 21). Both are now repaired, the first by separating the two cases and the second by replacing the argument with the multiplicative identity of Theorem 15. We record the episode because it illustrates the value of exhaustive verification against exact counts, and of machine-checking the statements most likely to be wrong: the check of Table 3 was designed to catch implementation errors, and instead caught a theorem-level gap.

Relation to earlier work. For base 10 the exclusion class $g \equiv 20 \pmod{40}$ and a measurement of $\delta(10)$ were obtained by the author in earlier work; the present paper subsumes that case, since $d = 10$ gives $D = 40 = 5 \cdot 8$ and Theorem 11 returns the single exclusion class $g \equiv 20$.

Garcia–Kahoro–Luca [3] study primitive-root bias for twin primes, i.e. the single gap $g = 2$, comparing $\varphi(p-1)$ with $\varphi(p+1)$. It is worth being explicit that our law says nothing about twin primes for the bases one might first consider. By Table 1, $g = 2$ is reversing only if some component reverses at 2 and none is indeterminate there; the component -4 reverses at $g \equiv 2 \pmod 4$ and the component -3 reverses at $g \not\equiv 0 \pmod 3$, but a component q^* with $q \geq 5$ is indeterminate at $g = 2$, and the component ± 8 is indeterminate at $g \equiv 2 \pmod 8$. For $d = 3$, with $f = 12$ and components $\{-3, -4\}$, both components reverse at $g = 2$, so the count of reversing components is even and $g \equiv 2$ is *preserving*, not reversing — as Table 2 records. Concretely, 3 is a primitive root of both 5 and 7, a twin pair at gap 2, so $g = 2$ is certainly not an exclusion class for base 3, and no reversing mechanism excludes gap 2 for any base.

The inadmissibility mechanism, however, does. Base 5 has conductor 5 and exclusion classes $g \equiv 2, 3 \pmod 5$ by Theorem 17, and $2 \equiv 2 \pmod 5$; hence Theorem 8's conclusion holds at gap 2 for base 5 by Definition 7. In other words, *no pair of twin primes* $p, p+2$ *can have* 5 *as a primitive root of both*, and the same applies to every base with squarefree part 5, such as $20, 45$ and 80. We verified this against the 3,804 twin-prime pairs below 4×10^5: none has 5 as a primitive root of both members, whereas 953 of them have 3 as a primitive root of both. This appears to be the only gap-2 statement our classification yields.

Pollack [14] constructs, under GRH, infinitely many bounded-gap prime pairs sharing a given primitive root, and Baker–Pollack [1] obtain an unconditional result for primitive roots drawn from a large set of primes. Our results add a base-specific constraint on where such configurations can occur: a pair sharing the primitive root a must avoid every exclusion class of a, and for $a = 3$ this excludes three of the six even gap classes modulo 12.

Future work. Three directions suggest themselves. First, the channel decomposition displayed after Observation 24 should be carried out: a derivation from the singular series of [6] together with Hooley's density computation [8], summed over the $\varphi(f)$ residue channels, would replace an empirical regularity with a formula and settle the exponent question. Second, the behaviour of $\delta(a)$ as x grows should be measured for several bases at once. Extending the present computation to 10^{12} is not immediate: the implementation fixes a trial-division bound of 10^5, which suffices to factor $p-1$ for $p \leq 10^{10}$ but not beyond, so a run at 10^{12} requires raising that bound to 10^6, with the corresponding increase in sieve memory and per-prime cost, and we have not measured the resulting runtime. We therefore record it as future work rather than as a routine extension. Third, the classification invites an analogue for higher-order obstructions: membership in A_a also requires a to be a non-ℓ-th-power residue for every $\ell \mid p-1$, and the cubic case in particular should produce exclusion laws governed by conductors of cubic characters, with $\mathbb{Q}(\sqrt{-3})$ playing a distinguished role. Such an analysis would also narrow the gap, noted in Section 1, between the quadratic exclusion classes classified here and the set of gap classes at which doubly-Artin pairs genuinely fail to occur.

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Use of generative AI. The author used a large-language-model AI assistant in the preparation of this work: for drafting and revising exposition, for writing and debugging the sieve and analysis programs, for exploratory discussion of the classification, and for the Lean formalisation described in Section 4. The present revision was also prepared with such assistance. Several specific interventions should be recorded in the interest of transparency. An initial machine-generated form of Table 1 incorrectly classified the conductor-8 component as preserving at $g \equiv 2 \pmod 8$; the error was detected by exhaustive computational comparison and corrected before any proof was written. The hypothesis refuted in Observation 23 was originally proposed, with a plausible heuristic, in the course of that assisted exploration; it was tested and rejected on the evidence. Conversely, three errors that survived into a previous version of this paper — the missing hypothesis in Theorem 17, the invalid composite case of Corollary 20, and a false assertion about twin primes in the discussion — were identified during a subsequent assisted audit and are corrected here; the counting identity that repairs the second is machine-checked, as are the $d = 5$ and $d = 13$ cases of the first; the general-d count and the composite dichotomy remain pen-and-paper proofs. The author is solely responsible for the content of this paper, including any remaining errors.

Declaration of interest

The author reports there are no competing interests to declare.

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Data availability

The code and results supporting this paper are available at https://github.com/jpbald93/consecutive-artin, including the segmented sieve and multi-base correlation program, the classification and verification scripts, the exact contingency tables underlying Tables 3 and 4, scripts reproducing Figure 1, and the Lean 4 development described in Section 4 together with the script that checks its axiom dependencies. The computation reported here can be reproduced from the repository on a single core; rebuilding the Lean development additionally requires a `mathlib` binary cache.

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